

Flock ChatApp - System Architecture

Full-Stack Multi-Tenant Real-Time Collaboration Platform

' PRODUCTION GRADE • MULTI-TENANT ENTERPRISE ARCHITECTURE

1. Executive Overview

Flock ChatApp is a full-stack, enterprise-grade real-time team collaboration platform built with React, Node.js/Express, Socket.IO, and MongoDB. It provides multi-tenant workspace isolation, 1-on-1 direct messaging, team channels, real-time presence, audit logging, to-do management, and role-based administrative control.

2. High-Level Architecture Layers

- **Frontend Client (React 18 + Vite):** Single Page Application (SPA) designed with a modular component architecture, centralized SVG icon system, responsive mobile rail/drawer system, and contextual state providers (AuthContext, SocketContext).
- **API & Real-Time Gateway (Express + Socket.IO):** Dual-protocol communication server managing REST API endpoints alongside low-latency bi-directional WebSocket events for instant messaging, presence tracking, and notifications.
- **Data & Storage Tier (MongoDB + Mongoose):** Multi-tenant database schema with strict indexing, compound organizational uniqueness constraints, and transactional consistency across 13 Mongoose models.

Core Technology Stack

- !' Frontend: React 18, Vite, Vanilla CSS Design System, Lucide-style SVG icons
- !' Backend: Node.js, Express.js (MVC Pattern), Socket.IO 4.x Gateway
- !' Database: MongoDB 6.x / 7.x, Mongoose ODM with custom indexes
- !' Authentication: Stateless JSON Web Tokens (JWT) & bcrypt password hashing
- !' File Handling: Multer disk storage statically served under /uploads

3. Multi-Tenant Organizational Data Architecture

Flock implements scoped organizational multi-tenancy where every workspace acts as a self-contained collaborative boundary:

- **Organization Isolation:** Every channel, member association, audit event, and join request is bound by organization ID. Users can belong to multiple companies and switch active contexts dynamically.
- **Compound Unique Constraints:** Channel names are unique per organization (`{ organization: 1, name: 1 }`), enabling distinct organizations to maintain standardized channel names (e.g. `#general`, `#announcements`) without conflict.
- **Role-Based Access Control (RBAC):** Granular permission tiering: System Admin, Organization Administrator, and Member with scoped capabilities for channel creation, user moderation, and audit visibility.

4. Key MongoDB Collections & Schemas

- **organizations:** Company metadata, slug, creator ID, and custom organizational policies (`requireJoinApproval`, `allowPublicChannels`).
- **memberships:** Maps users to organizations with assigned role (`admin/member`) and status (`active/suspended`).
- **channels:** Public and private team channels, member rosters, and organizational reference.
- **conversations & messages:** Direct 1-on-1 chats and channel messages with support for reactions, file attachments, and pinned/saved flags.
- **joinrequests:** Join requests lifecycle (`pending -> approved/rejected`) with reviewer tracking and real-time Socket.IO dispatch.
- **todos & auditlogs:** Task delegation and immutable audit trail records for compliance and administrative scanning.

5. Real-Time Socket.IO Pipeline

The real-time layer leverages room-based multiplexing to dispatch events with sub-50ms latency:

- **Presence & Online Tracking:** Maintains an in-memory Map of active socket IDs to user IDs. Broadcasts user:online and user:offline events on connect/disconnect.
- **Room Scoping:** Users automatically join personal rooms (user:userId) for direct notifications and specific channel/conversation rooms upon viewing.
- **Live Messaging & Typing:** Events message:new, typing:start, and typing:stop provide live feedback without polling or database stress.
- **Instant Join Request Flow:** Join requests emit join_request:new to the admin room in real-time. Upon approval, join_request:approved is sent directly to the applicant.

6. Responsive Frontend & Mobile Design

- **Desktop Layout:** Three-column workspace layout: Primary Navigation Rail -> Sub-panel List -> Main Communication Stage.
- **Mobile Experience:** Fixed bottom navigation bar for high-frequency workflows (Chats, Channels, To-Dos, Team Directory), animated off-canvas drawer for secondary tools, and modals hoisted to root viewport scope.
- **Exact Search Engine:** Backend-optimized regex exact-matching prevents partial over-matching and hides company lists on initial empty search.

7. Security & Non-Functional Highlights

- **Zero Stale Renders:** Hoisted modal views prevent CSS transform clipping bugs during mobile navigation drawer transitions.
- **JWT Security:** HTTP authorization headers verified on every protected route with payload-embedded user IDs.
- **Audit Compliance:** Role changes, company modifications, and channel creations generate timestamped audit records.